SQL Basics – Data Analytics Practice Assignment

Scenario: Retail Sales Analysis. A retail store wants to analyse its sales data using SQL.

Questions

1. Create a database named retail_db.

2. Select/use the retail_db database.

3. Create a table named sales with the following columns:

Column	Description
sale_id	Unique ID for each sale
product_name	Name of the product
category	Product category
quantity	Number of units sold
price	Price of one unit
customer_city	City of the customer
payment_mode	Payment method
sales_person	Name of salesperson

4. Insert all the given sales records into the sales table.

```
INSERT INTO sales
(sale_id, product_name, category, quantity, price, customer_city, payment_mode, sales_person)
VALUES
(1, 'Laptop', 'Electronics', 2, 55000, 'Bangalore', 'UPI', 'Anil'),
(2, 'Mouse', 'Electronics', 5, 800, 'Mysore', 'Cash', 'Bhumika'),
(3, 'Keyboard', 'Electronics', 3, 1500, 'Bangalore', 'Card', 'Chetan'),
(4, 'Chair', 'Furniture', 4, 3500, 'Mangalore', 'UPI', 'Anil'),
(5, 'Table', 'Furniture', 2, 7000, 'Bangalore', 'Card', 'Dhruv'),
(6, 'Notebook', 'Stationery', 10, 80, 'Mysore', 'Cash', 'Bhumika'),
(7, 'Pen', 'Stationery', 20, 30, 'Bangalore', 'UPI', 'Chetan'),
(8, 'Printer', 'Electronics', 1, 18000, 'Mangalore', 'Card', 'Dhruv'),
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(9, 'Desk Lamp', 'Furniture', 3, 1200, 'Mysore', 'UPI', 'Anil'),

(10, 'Headphones', 'Electronics', 4, 2500, 'Bangalore', 'Cash', 'Bhumika'),

(11, 'Bookshelf', 'Furniture', 1, 9000, 'Mangalore', 'Card', 'Chetan'),

(12, 'Marker', 'Stationery', 8, 60, 'Bangalore', 'Cash', 'Dhruv');

5. Display all records from the sales table.

6. Display only:

- product_name
- category
- price

7. Display all sales where the product category is **Electronics**.

8. Display all sales made by salesperson **Anil**.

9. Display all sales where the quantity sold is greater than **3**.

10. Display all sales where the price is greater than **5,000**.

11. Display all sales made in **Bangalore**.

12. Display the product name, quantity and price of products where the payment mode is **UPI**.

13. Display Electronics products sold in Bangalore.

14. Display sales where the quantity is greater than 3 **and** the payment mode is UPI.

15. Display sales where the category is Furniture **and** the city is Mangalore.

16. Display sales made either in Bangalore **or** Mysore.

17. Display Electronics products where the price is greater than 10,000.

18. Display sales made by Anil or Bhumika.

19. Display products whose price is between 1,000 and 10,000.

20. Display products whose quantity is between 2 and 5.

21. Display sales from Bangalore, Mysore and Mangalore using IN.

22. Display sales where the payment mode is either Cash or Card.

23. Display Furniture and Electronics products using IN.

24. Find the total number of sales records.

25. Find the total quantity of products sold.

26. Find the highest product price.

27. Find the lowest product price.

28. Find the average product price.

29. Find the total quantity of Electronics products sold.

30. Find the number of products sold using UPI.

31. Find the average price of Furniture products.

32. Find the number of sales made in each city.

33. Find the total quantity sold in each city.

34. Find the average product price for each category.

35. Find the highest product price in each category.

36. Find the lowest product price in each category.

37. Find the number of sales handled by each salesperson.

38. Find the total quantity sold by each salesperson.

39. Find the number of sales for each payment mode.

40. Find the total quantity sold for each category.

41. Display cities where more than **2 sales** were made.

42. Display categories having more than **3 sales**.

43. Display salespersons who handled more than **2 sales**.

44. Display categories where the average product price is greater than **2,000**.

45. Display cities where the total quantity sold is greater than **10**.

46. Display payment modes used for more than **3 sales**.

47. Display all products from highest price to lowest price.

48. Display all products from lowest price to highest price.

49. Display products alphabetically according to their names.

50. Display categories and their average product price, starting from the highest average price.

51. Find the total quantity sold in each city and display the cities from highest quantity to lowest quantity.

52. Find the average price of each category and display only categories whose average price is greater than 2,000.

53. Find the number of sales made by each salesperson and display only salespersons who made more than 2 sales.

54. Find the total quantity sold for each category and arrange the categories from highest to lowest quantity.

55. Find the average product price for each city and display the cities from highest average price to lowest average price.

56. Find the number of sales for each payment mode and display only payment modes used more than 3 times.

57. Find the total quantity of Electronics products sold in each city.

58. Find the average price of products sold in Bangalore for each category.

59. Find the total quantity sold by each salesperson and display only salespersons whose total quantity is greater than 10.

60. Find the number of sales for each category, but display only categories having more than 3 sales, and arrange them from highest number of sales to lowest.